# Aswanth Bindu Ajith

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## 1PROFESSIONAL SUMMARY

M.Sc. student in High Performance Computing and Quantum Computing with a Mechanical Engineering foundation. Comfortable building quantum algorithms, software engineering solutions from low-level numerical code to production web features. Reduced simulation runtime by 40

## 2TECHNICAL SKILLS

Quantum Computing: Qiskit, Cirq (basic), Quantum optimization algorithms, Variational quantum eigensolver (VQE)

Programming Languages: Python (Advanced, 4+ years), C/C++, JavaScript, Bash/Shell scripting, HTML/CSS

HPC Parallel Computing: MPI (basic), OpenMP (basic), GPU-aware computing concepts, Cluster job scheduling (SLURM basics), Performance profiling

Modeling Simulation: Finite difference methods, Computational fluid dynamics (basics), Thermal simulation, Multi-physics modeling

## 3PROFESSIONAL EXPERIENCE

**Working Student Software Developer** — *Germany*

*No Border (E-commerce)* — Feb 2025 Present

- Shipped 5+ frontend features in a 2-week sprint cycle, reducing average ticket age by 30
- Debugged and resolved 20+ production bugs across the React/Vue stack, cutting reported error rates by 25
- Wrote unit tests for 3 core modules, pushing test coverage from 45
- Collaborated with 4-person cross-functional team (design, backend, QA) using GitHub Flow and daily standups

## 4EDUCATION

**M.Sc. High Performance Computing and Quantum Computing** — 2024 Present

Deggendorf Institute of Technology (DIT), Germany — Current Average: 2.6

*Relevant coursework:* Parallel Distributed Computing, Numerical Optimization, Quantum Algorithms, Scientific Software Engineering, High-Dimensional Data Analysis

*Highlights:* Optimization Methods (1.7); System Design Applications (1.7); Software Engineering (2.3); Advanced Mathematics (2.3)

**B.Tech Mechanical Engineering** — 2019 2023

APJ Abdul Kalam Technological University, India — CGPA: 6.48

## 5PROJECTS

**Large-Scale Computational Simulation Data Analysis**

- Built a Python simulation engine processing 30,000+ interacting entities using vectorized NumPy operations
- Reduced runtime from 12 minutes to 7 minutes (40
- Identified 3 non-obvious data correlations that shifted the team's modeling assumptions
- Packaged results into an 18-page LaTeX report with reproducible Jupyter notebooks for peer review
- Key metrics: 30,000+ entities, 40

**Computing Architecture Optimization Methods**

- Implemented gradient descent and genetic algorithms from scratch to optimize a 50-dimensional engineering cost function
- Benchmarked CPU vs. GPU (CUDA) performance on matrix operations, documenting 8x speedup for large matrices
- Profiled memory bottlenecks using Python's cProfile and reduced peak RAM usage by 35

- Presented findings to a 25-person seminar, receiving top marks on the technical report
- Key metrics: 50-dimensional optimization, 8x GPU speedup, 35

**Thermal Analysis of Battery Modules**

- Modeled heat transfer across a 12-cell lithium-ion battery pack using finite difference methods in Python
- Simulated 8 operating scenarios, identifying a thermal runaway risk at >45řC under sustained 3C discharge
- Proposed a revised cooling channel geometry that improved peak temperature uniformity by 18
- Co-authored a technical report cited by 2 subsequent student projects in the same lab
- Key metrics: 12-cell pack, 8 scenarios, 18

## 6RESEARCH INTERESTS

Quantum Computing, Quantum Algorithms, High Performance Computing, Parallel and Distributed Systems

## 7LANGUAGES

English (Professional Working Proficiency), Malayalam (Native), German (A1 (Beginner))